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Can you update me a bit about like right now and then we'll talk about like direction of your project a bit later, but could you update me about like your progress with EzXim? Yeah, I have the EzXim up and running. I'm trying to collect some data. I have two problems that I'm currently dealing with.

One of them is that we, for the Barrett simulation that they handed to us, he doesn't really control the gripper. He have a fixed range of closure for the gripper. It's 55%, I guess, 0.55. And that is working for the example that he has.

He has a cylinder with diameter of 26. And when we close the gripper with that range, we get to a point that we get to read the tactile data. But for other different, if we replace the object with something bigger or smaller, we are still going to have the fixed range for the gripper.

So we need to adopt that. What he told me that he said in the lab, he's trying to find the optimum range of closure for the fingers for each object that he has. And then he would apply the number in his simulation.

So what I'm trying to do, I'm trying to read the tactile data and find a threshold that I realize that contact started and then stop the fingers for every object. So this is the thing that I'm working on. And the other thing is that because on the data collection process, we want to know the exact position of the tactile pad in relation with the object.

We need to know the offset between the end effector and the pad at the closure, at the closure statement. And it would be different for different objects. Based on the diameter and width of the object, we would have the fingers at different offset in relation with the end effector.

And I need to be able to find that offset. So I would know exactly the position of the pad, like sensor pad on the object. So I can reconstruct the big tactile map.

So these are the two things that I'm working on right now. And also, I'm trying to read some papers regarding the construction of the big map. When we have like, I have 100 points of tactile data.

And so far, I'm doing just a simple averaging of the when I have an overlap of the taxels, I'm trying to just get an average of all the taxels and replace that with the for the big tactile map. Sometimes it works. But most of the time, it doesn't give me a complete picture.

It's not clear. Like if I am if I'm grasping a cylinder doesn't give me the cylinder. It's just scattered points.

I have pressure at different points. So I was just With the digital twin? Yeah. Yeah, with the digital.

Yeah, with the exact same. Okay. And so why does having contact with Hi, Jean-Philippe.

Yeah. Hey, Jean-Philippe, ça va? Ouais, ça va bien, toi? Ouais, super. Super, super.

Laissez-moi deux secondes. Je vais starter une affaire. Ça va reprendre 20 minutes de cloud.

Laissez-moi. Okay. Parfait.

Merci, tout le monde. So, yeah. Okay.

So, but do you understand? Okay. Maybe, Koush, let's set that aside, the technical part. While we have Jean-Philippe, let's go back to what we talked about, like defining the project.

So, Jean-Philippe, well, I hope you had a great wedding and also a vacation that came after. Yes, yes. Yes, it was great.

Yes, it was. Cool, cool, cool. Well, let's, so while you were away, Koush, we're at the point to start his last thesis chapter or paper.

Yeah. And so, we talk and just want to start with a summary of what I said and the reason why I said that. And then, he told me, you suggested also the thing around that and just wanted to, like, we discuss and we agree on the point and on the plan.

So, I started with two things. So, Verit released, like, a new version of the tactile sensor, Digital Twin. I thought it, for Koush, background in Digital Twin is now more and more part of the things of robotic field.

I thought it was good for him to play on that. Well, I saw that also as a way to actually do things we should have done from the start that we could not due to experimental limitations. So, the main point is, like, one of the weaknesses of the first paper is doing the imagination of, re-imagination of the contact outside of the grasp area.

So, in one of the way we did is using, like, convexity and non-convexity, like, for flat contact, cylindrical contact. And so, and we had that classifier that was mapping to a sphere, cylinder, and all this stuff. And then, having an algorithm that just expanded.

Then, we sent it to the classifier to say, is that better grasp or not? And, the point was, like, we could probably, having thousands and thousands of data of object, different object, we could probably extrapolate what it is. And, one of the idea, I don't know if you saw that, it was very impressive, it did just show that with only 6% of the pixel randomly dispersed in an image, you are able to reconstruct very accurately the image. Six pixels? With 6%, 6%.

Okay. So. But still, yeah.

Maybe eight. Okay. So, and my thought is, like, to build that, you need a lot of data.

But, now, with digital twin, we could have a lot of data. So, yeah, well, so, that's what we did. So, the idea, if I got that right, is, like, to skip the prototypical shape recognition and just go for having a lot of data and trying to do grasping assessment.

Yeah. So, that last chapter would be pretty well aligned with the rest of the thesis. It's more like, now, with digital twin, we're able to gather enough data to create a generative AI that actually give you this reimagination.

And, that would be probably more, since it's generative AI, would be more like a reimagination of the object outside of it instead of just determining this model of the cutback. Yes. So, the main part is the rest.

We find that, like, in the recent paper, how to go to the optimum as fast as we can or less simple. So, that was the idea. I'm open, totally, to other things.

And so, it's, in any case, we haven't done a lot so far, except, like, I have pushed to get familiar with the exact same together data. My understanding right now is there's still a lot of limitation with what Barrett has been doing, and that could create, actually, issue to gather data. But, that's one point we could talk.

But, before jumping onto the technical, Kourosh told me also you had ideas. So, like I said, I'm open to... But, Kourosh, last time we met, I thought you said you started gathering data using Barrett's plugin. And, I would be curious, just for my own knowledge, to know what's the limitations from your side.

Yeah, I started the data collection part, but I don't have the data yet. The two things that I'm currently working on is that I have an example from Barrett's simulation that is a touch cylinder. I think you're familiar with that.

And, that cylinder is, like, the diameter is 26 millimeters. And, based on that diameter, Barrett has a fixed number for the range of the gripper. How much he wants the gripper to close around that object until we get the tactile data.

But, if we change the object, Barrett told me that he had the exact same replica of the object in the lab that he would go and experiment with the actual gripper. And, he would find the range of the closure at which he start to read the tactile data. So, I need to have a pipeline that where I close the gripper, I can, like, the simulation can detect where the tactile data starts to making sense, and then we stop.

So, this is one thing that I don't think it's a... I can do that, I think. So, I'm working on that. And,

the other thing is that, again, it depends on the, like, how much is the offset of the sensor path from the end effector of the robot that we can read, based on the object that we are grasping.

So, if we have a bigger object, it would stay upper, like in z-axis. And, if we have a thin object, it would... So, I need to find the exact position of the sensor path in relation to the object surface in order to reconstruct the big map. Okay.

So, just one question. So, what would be different than, like, previous work where we were trying to do using real data, I agree, but where we were trying to do, like, grasping assessment? Like, the work from Dean Cockburn, back in the days, where we'd use, like, sparse coding, but it doesn't really matter the method, but your novelty is just, like, trying to generate the contour of the tactile map, right? But, how that would be an added value compared to just trying grasping assessment directly from the tactile input, the raw tactile input? But, well, I don't think it changed. I don't think it changed anything on the grasping assessment part, which we're seeing right now.

It's more looking at... It's really more generating the tactile outside of the grasp to find, like... I don't know. So, the question is more, like, given a large set of synthetic data, can we imagine the contour of a real tactile map? Is that...? Yeah, it's really more, like, you have this tactile sensor, and then you can extrapolate what should be the contact if you move by, like, inside of a centimeter all around, two centimeters. So, it's really, like, extrapolate what is outside of the grip, and that hasn't been done directly.

Like, right now, we're doing it mapping through prototyping shape, and then using the terminus model to say, yeah, it's a cylinder, so outside, it's going to look like that. If we extend the frame, it looks like that. So, but at the end, it's still using the same classifier of Jennifer right now uses.

Yeah, yeah. Then, you take this extrapolated contact, and you feed it to the grasp. But the main difference would be that we skip the shape recognition part, right? Yeah, which was kind of, it worked, but it's pretty limitation.

So, first, you depend on the classifier for the shape, and if you get it wrong, all extrapolation is wrong. But also, it kind of saying that most of the contact map to those three different shape, which we could argue is it cover a lot. So, if we get it right with the cylinder, and the sphere, and this thing, and getting it right, by the way, digital could help us, because, like, we took a couple of cylinder, a couple of sphere to do things with digital twin, we could actually increase and make that classifier better.

But the whole point is, like, we move to generative AI to generate that extended map, and then we just use whatever we have as a classifier to find the optimal point of grasp. Okay, and sorry, I'm asking a lot of questions, but it's just for me to make sense of all of this. Follow this.

So, like, earlier, we were kind of using a model-based approach. It's not a pure model-based, because we use AI to classify into three possible shapes. But then it was modeled by AI.

Yeah. So, now we're using generative AI, and that would be the main contribution, and then we use that as a way to improve grasping assessment. Yeah, totally.

Okay. And I was actually thinking, also, as we will do that, to benchmark with image model of, like, we build the classifier, but I would be curious to end up with something I would see in the paper. We send those image to – JGPT is now back on top on the image model, but Gemini 2.5 was probably, like, the NanoBanana2 was, like, probably the better.

I would be curious to benchmark with an image model specifically to say, hey, what is the upside, like, try to think of this image and increase, just extrapolate the contact one centimeter around, like, given the distance. That is something I would be curious to see what you think. Okay.

Okay. Excellent. All right.

That's interesting. Also, one thing I might be able to help, Kourosh, is right now, Berit, just at the beginning of the week, we resubmitted the paper, and he was very busy with that. Right now, he's working on moving to Newton.

I don't know if you know about that, Kourosh, but I know Vincent, you know about this. No, no, but I know him because my team at the Methodist Academy had a meeting with Berit, actually, yesterday, and we now have one of the best guys we have internally that is on his XM. So, I think he's probably going to be also able – so, he fixed the issue with the two-finger gripper that – but, yeah.

And he shifted to Newton right now instead of physics. Yes, yes. So, there's a solver, Kourosh, inside of it that solves the physics part of the things.

It's really the part – I've been explaining that – is NVIDIA come from gaming, so they're pretty good at doing, like, visual part, but the physical simulation sucks. Like, for example, closed-loop kinematics, they don't do it. It's really, like, a nightmare to simulate closed-loop kinematics, while, like, MSC add-ins, which was shitty on the visual part, but could do it in 2000, 2001.

So, it's really, like, they struggle on the physics part, but they're very good at the visualization part. And he tried to bridge the gap in Newton, his initiative toward this, and it's – yeah, they're very dynamic around Newton. There's a new feature almost every week on it.

Yeah, because it's true that, like, the whole sensor – back in the days, what we wanted to do as a contribution is just to show that we were able to generate realistic data. But we are just trying now to really have it in a complete robotic setup, and it was very difficult using physics because – well, anyway, for different reasons. And now we're trying to do it and do some learning for another task, Gaurash.

So, you're not working on the same thing, of course, but we're kind of aiming towards the same

direction. I know during my vacations, I'm not completely caught up to that, but I know that Berit also, with Robotic, they also wanted to have something that would work on a real robot with a real gripper. So, I think right now we could try to work with Berit to make you a setup that would generate the data as soon as possible.

Yeah, JB, are you going to be to IROS? No, no, no. This year or next year? Yeah, this year. No, no, no.

Because we actually took a boot and we're planning to have part of it that's going to feature some demo around exact sims. And if things work on your side, there's already discussion to feature Berit's work, even if it's not a point that it fully worked with what you did. Okay.

Where is it this year? So, it's in Pittsburgh. Pittsburgh, okay. There's also a workshop, Mark is part of it, of the organizing, but it's a workshop on bringing tactile sensing into, I could send you the link, and a very good name there, and bringing tactile sensing in current AI.

Because most of the current AI is all VLA. And it was all VLA. There's a strong shift.

Partially, I've been told in Silicon Valley that now VCs start to get scared of VLA because they realize that vision cannot be the only pipeline for speed of efficiency reasons. But also, at the end, it failed to capture the physics behind pixels, like what happened at the touch. So, there's a strong shift to touch, so there's actually three to four workshops on that topic, but it's one with a lot of good names, and Mark is on that committee, too.

So, you might be interested to go there since it's not that far. Yeah, yeah. I didn't realize it was in Pittsburgh.

So, yeah, I will check if that's possible, because I teach two courses this fall, but I will see if that's possible. Yeah. That would be great.

Have you seen, by the way, have you seen the demo we did on Frank on The Rock when we presented also the digital twin of Beirut? No, no. Let me just show a video, and it will help you to understand a bit more what we have in mind for defining those videos. Yeah, so, yeah.

Yeah, music is. So, it's a classifier that has been trained to recognize shape, and you might see it as like virtually the cube. It's feeded sometime in a way that it doesn't look like a cube.

I think it's going to do the cube soon. Okay, yeah. Yeah, even the cube, yeah, and that axis is still being able to recognize it.

It's used with few-shot learning. So, we have this demo working. So, one plan was to use Beirut classifier, Beirut digital twin, and like span it, like having ten of it in the same screen, and they all grab shape, and they get our data.

Okay, okay, okay. Yeah. To show, the main point is to show that with digital twin, you can speed up training, because honestly, since it's few-shot learning, just few part, the real thing wasn't that hard.

It took like one night for the guy that did the demo to gather all the data, but the point is like showing the example, like you can use digital twin, and then get. Yes, yes, I see the point. Yeah, that would be great.

It's very cool. And maybe I can show you another one that was cool too. That's simple and very efficient, the video that you showed.

Oh, by the way, this is, okay, I'm not going to shift inside, but now with the tactile, and partially now that we have back the dynamic, and on this gripper that's 60% speed, but now we're able to do the 200% speed to grasp raspberry without breaking it. Nice. So, and it's really dynamic doing it, but that's not the other demo I wanted to show in this one.

The music is horrible. So, that's an application we hear a lot. So, I'm surprised to see the interest in tactile sensor.

The amount that we sell, we're always backorder and crazy. And when it's frontier model lab, it's mostly they want to do their own shit with it, but for more applicative people, one of the top application is recognizing the angle at which you grasp. The pose? The pose of the... Yeah, the pose of the sensor actually push the algorithm that I sent you for recognizing the vectors which we use as a base here.

But on top of that, we did a couple of, a bit of machine learning to improve the accuracy. Actually, as human, we're not able to do that stack. Like the first five is not that hard, but none were really able to do the full stack like robot is doing right now.

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So, Kourosh, right now you're trying just to gather the data, that's kind of the bottleneck? Yeah, I can say that, yeah, I'm trying to get the data and then the next step is right after that where I try to stitch all the data together to make the big map and then relate each of my single grasp to that big map, so I can start training a model, a model that the input is just a single grasp and the output is the big reconstructed map and, yeah, and the... Sorry, so you will be stitching, like, synthetic tactile maps that will be used as ground truths in your simulation? Yeah, and we wanted to, we talked with Vansan about that, that we can use real data and synthetic data, like from the digital twin at the same time and maybe we'll have, we'll see how the model is performing, the one which is purely synthetic data and the other one which is coming from the combination of the two, so I'm going to have to collect some real data as well and see how they are different from the synthetic data. Yeah, the stitching part is not that easy because there's an overlap, you're not going to move in a way that there's no overlap, so, but, yeah, and there's part, for example, well, it's what you expect, yeah, the point, yeah, the stitching, when you move around in the object, it's not necessarily going to map the shape of the object, which really happens when, like, just think about, you grab a ball and then you move, it's just more like the point of contact will just move toward the tip, for example, if you move out of the ball and you just move, but, so, if you stitch them, it looks like a line, so it could look like a cylinder, but the point is, it's really the contact you would have at that point if you move outside of it. So, yeah, that's a good point, so I guess the approach to do that is kind of, you have to determine the tactile imprint and then, so from the contact patterns, you try to process the tactile imprint and stitch the tactile imprints as you move? Yeah, yeah, yeah.

Yeah, I agree, it's not an easy problem, it's easy to understand, but when you will do it, I think it's more difficult than it seems. Yeah, I think the simplest way is to just average where they overlap, like, if I have a tag cell where I have three frames, I can have it, but it's not, I don't think averaging is the answer. No, but probably, but I don't want us to dive into fancy, too fancy algorithms, but I would think about ICP as well, iterative closest point, when you have, like, a deformation, like a tactile imprint, and just to stitch them properly, yeah, anyway, I would process probably that as well.

Sorry, I took notes in Google, do ICP, because I didn't know, and you suggest me in-sign clones. Yeah. Is it ICP you said? I used to listen to that band when I was a teenager.

Okay, I didn't know that, this one, but ICP. ICP is iterative closest point, it's probably the most used algorithm for stitching point clouds and stuff like that. But actually, thinking of it, it's not necessarily stitching we need, because, you know, let's think about, like, a contact, like, you grab a sphere, and it's perfectly aligned in the middle of the gripper.

Yeah. Right now, the algorithm we have, we have the deck, it's a sphere, and the extended image would still have been, like, no contact all around, and moving the frame in it would just have moved that point of contact around. Now, if we go and grab other points, like... I would reference, so this example, for example, with tactile imprints, the forms, if we generate the tactile imprints from the tactile maps, I would suppose that it stays almost redundant as you move the, because you reference in the reference frame of the robot or in inertial reference frame, so when the gripper moves, the tactile imprints stay stable if it's a sphere.

It would also, unless you change the angle, and you rotate around the center of the sphere, then it would just generate new, then you can start stitching, because it would generate new data. But if you're just touching the same contact point over and over, but moving the gripper in different, it would still, for me, locate the imprint in the same way. Yep.

Yep. So, that's where, for a point, in this case, not stitching, it's just, like, we get a different view, like, you keep the point, then just the frame that's switched. What happens if there's more complex contact? For me, sorry to insist, but it's still a stitching problem for me.

If you change a robot with still the same contact point, it would overlap completely, theoretically, the same way. It would just create a redundant, but still, it would be, it's like if you take two 3D LiDAR scans in the same room from the same location, you could stitch the two point clouds, even if they are exactly the same, or more or less. But, you know, the contribution of that, to me, is, like, when I'm thinking, and I'm just, I want to keep the, so, stitching, yeah, it's definitely going to be one challenge to this.

I would probably, and Koush, want to take some real data out of it, but I think it could be relatively simple, not necessarily having, like, we would rely mostly on simulation, but a bit of real data. And the other points, like I said, I would like to compare. So, probably, we could play with a mix of real and non-real data, and show that, like, the still twin of Beret can provide accurate, or not, like, helps understand it.

Then, I would actually benchmark with, like I said, image, large image model, to see if the, actually, we already have this knowledge encapsulated in them, and that we don't actually need to train something. My guess is, well, I don't know. I don't know.

I'm still, sometimes, very surprised when they come out, but, yeah. Wansun, you mentioned this benchmarking against, like, I believe you mentioned ChatGPT, or the other one from Google? Yeah. Is that? So, I'm sure that they can give us something back, if we give, like, if feeding the single tactile, but if they can do that, how is our work, like, unique, or how is it? I mean, what is our contribution? If it turns out that, yeah.

We will prove it, that they can, and that's a good contribution in itself. If they can do it, it's good. Because we will gather real data, and simulation data, but the simulation data, it's more, will they

render very well on the tactile sensor, but it will be good, like, what the data will generate will be a good, like, you move the robot, it sends here, you grasp, and, like, it should reflect very well, so that will be a data set of, hey, that is the actual extrapolation of the initial contact, so we'll take initial contact, and that is, like, the extended map of that contact.

Right. So, we will be able, so when we will get, so we will take just the initial contact you have, we'll feed it to a chat GPT 5.6 latest image model, and then we'll feed it to a Gemini 3.5 image model, and we will be able to get some extrapolation, and then we'll compare them, because we will have a ground truth, which is the data we have. Right.

And at the end, like, we will make a model that is supposed to be good at generating those patterns, so at the end, we'll be able to compare, again, the ground truth against our class of generative AI algorithm, and when we'll get that step, I will generate you, because it's not something you will be able to do it in chat GPT, so we will, like, I will generate you a key to be able to just, like, send the image, get the image back in the pipeline. Oh, okay. It's going to be very simple.

That part is going to be simple. Just take time to take, like, 60 seconds by contact, and by the way, it's one thing, even if they would match, I think the large language model is too complex that it takes the same time to generate that than generate you a picture of you with a six pack on a beach with a lot of people. So, yeah, the point is real time image model is too large for what we do, but it would be interesting to see if that's intuitive enough for large language models.

I mean, that was one point of interest to me. Yeah, yeah, I think it's very... Yeah, no, I'm curious, too. We'll definitely check that in to see how it works, but we have to make the model, train it, and verify it first and see how it works.

Yeah. And yeah, I'm thinking about, I'm thinking about this, like, stitching problem all the time. I'm trying to find some papers and read on that and see how it was this idea of what if we have, like, weighted, weighted form of stitching for the center of the tactile and the edges.

So we treat the numbers on the center differently from the edges. And so, like, the problem that Vanson mentioned, that if we are touching a ball and we go a different position, at the end we get a line and not a, like, a centered focused contact. You're not supposed to get a line.

It's just a matter of putting the contact in the right reference frame, which should be an inertial reference frame. If it's a ball and you're moving the gripper around and always touching at the same place, it should just be redundant data. Yeah, you're right, you're right, you're right.

But if... But you're not touching at the same position all the time. I mean, what is the point? The point is, you know how much you move. You know you move one centimeter, like, in one axis.

Right. So you can keep the contact always aligned. So the point that moves one centimeter, you

can, like, re-center it by one centimeter.

Like, you can express the problem into the, let's say it's a ball right now, but you can express it in the ball reference frame. So you move relative to your initial contact in the case of the ball. And then whenever you move, you can offset it by the same amount and superpose.

So if there's something new, if it's a ball, usually you should just converge into superposing always the same frame. And that gives you pretty much how you move. But if there's something new that appears, it gives you more information of what's coming outside of it.

Is that how you would explain it? Yeah. Yeah, exactly. But still, it's, yeah, yeah, there's some challenge.

But as soon as you will get some picture, and maybe that could be a starting point, is just gather a lot of picture around with the coordinate of the reference frame, like, of the shift. So let's start. And we could do the ball example for, like, and if you want, and you're not up to that thing.

But do you think you could do it right now with the... I think so. Like, you could do with the cylinder. You said cylinder's working.

Like, you could grab cylinder different place, and you move in one direction, you move in the other direction. And then we could just start to express that image. Yeah.

Yeah, with the cylinder, yeah, definitely I can do that. I think I'm going to be, I'm aiming for, like, even sooner than a week. But I think in a week, I can have some good pictures and maybe try out different stitching methods to see which one is closer to the actual object.

Yep. So, yeah, I think I definitely... So the main point here, Krush and Jean-Philippe, and by the way, Jean-Philippe, if you have another idea around this, welcome. Actually, I couldn't have your input when we decided that, but... No, I think it's fine right now.

It's interesting where we're going, yeah. Okay, great. But, yeah, the main point, Krush, and we're already July 20.

Okay, so I would have expected at that point, since we started in June, it would be, like, up and running. I understand there's issue with various classifiers, but I want to be sure, because if we need to shift the whole idea of doing in virtual was, like, that you get technology, you get more, like, familiar with digital twin, and then, like, a week or two to get up and running, and then it would be easy to have a computer running data and data and data, and then machine learning is not as hard as it used to be right now, just as a thought. Yeah.

Yeah, I know I maybe mis-estimated the thing, but I know that we were planned to have it by now, but I don't know, I'm trying really hard. What I want to make here, if you feel like it become, like, it will become a nightmare, the whole point of digital twin was to avoid real experiment, like,

to go faster, and I knew you would have a ramp up a few weeks on the beginning, but at the end, like, you could generate thousands and thousands of data pretty easily compared to experimental data and different shape and different size and all this stuff. So, but if you feel that it drift, just tell us at some point, or, like, and it's the reason why I think we should meet every week, just to be sure that you're on target.

Yeah, sure. No, no, no, I mean, let me just understand you clearly, what do you think the alternative is if I think, I mean, it takes a lot of time, or do you think we can go back to the, like, a real robot and do it that way? Like, go back to master student. No, no, for real, no, no, no, for real, I don't know.

We need to think of something else. If you think, like, you're not going to be able to get digital imprint, like, tactile signal in the digital, we need to think of something else. Because on experiment, I think doing this project purely on experimental data would be incredibly hard, because I don't think here it's going to be a matter of hundreds of data, but it's going to be, like, thousands and thousands and thousands of data to be able to generate this.

And personally, as just, like, the robots will need to grasp and then to move around and, yeah. No, I mean, for every part of the project, I mean, the first two paper and here, I'm learning as I go. These are not things that I know already, but I'm, I can say that I'm confident and I'm, I want to do it.

No, it's, I, you know, my plan was to finish by September, by fall semester, but worst case, maybe I just register for the fall semester as well to be able to finish and wrap the whole thing up, if that's okay with you. So, let's try now to generate, what I would suggest is first, if cylinder is working, try to generate those things so we can work on this teaching part. Okay, so, point one, and try to see what's your limitation.

So, are you going to be able to have different object? And, you know, I think, and I'm sure you fully understand there, but you said that right now, there is, it's worked with pretty fine, like, close the gripper by pretty fine amount and then whenever you close them, like, I don't know, it's close to five centimeters, so we work with a cylinder of that size and then you get the, in this case, we could generate object that has this exact size. Yeah, all right. It wasn't straining to, like, particularly on cylinder, like, the curvature we can have, so, yeah, that's not ideal, but I think you said it is, like, couple of pretty fine size.

Yeah, that's what Barry told me. He said he would go in the lab and play with the actual gripper to find the optimized range for the gripper. I can do the same or keep the size at least in the same range of different shapes.

Like, if I have a square or cuboid, I can have the same widths as the diameter of the cylinder, you know what I mean? So, still, I get to pick up some tactile data upon the closure of the gripper, but I, yeah, I'm working on these two, the offset and how much I want the gripper to close, but as long

as I have the same cylinder, I can just skip this part for now and go ahead and see if I can work on stitching and collect some meaningful data with a good reference. Okay, with the cylinder, if it works, there's still, like, automating the pipeline to gather all the contact around the image and reconstruct it, so that's one thing, but it's kind of useless also if, at the end, we're not able to do, like, a hundred of different objects. I understand.

So, point is, you need also to investigate, will you be able to do other objects? Yeah, yeah, I'm gonna look into that. I think I, I mean, I'm gonna be able to do that. I'll manage to do that, though, I mean, this is an exciting project as the, like, a final wrap of the project, wrap-up of the whole PhD thing.

I'm excited about this, I'm, and I think I will find a way to do that, like, the challenging part is the stitching, collecting meaningful data for different objects, and this is the whole point, and then we'll see how the model is performing with the, and real data, and I think real data might not be a big problem, it takes some time, but I have the pipeline, I did it before for different objects, so I can collect real data later on, and combine it, but yeah, so. And you know, GP for the stitching, if you remind me, Euroscope, we have in the lab that is near cable right now, it's exactly that, he, like, the frame was, when you were doing stitching, the stitching was kind of a, the size of a tactile sensor, and then it just moved, and you were seeing the stitching occurring. Yeah.

So, yeah, okay. But, I don't know if it's normal or not, but I'm extremely stressed these days, I mean, like, I'm working on my computer, and I would just think that only for a second, I wasn't this stressed, I would accomplish more, but I see that you're obviously making joke about the master thing, and, but it's just really, I mean. No, no, no, no, it's not an option, I mean, it's really a joke, because the student going back to master is usually at the qualification exam here.

Yeah, no, but my friend and my wife even keep joking about this, and I'm saying. Yeah, my friend actually called me, and, no, but for real, no, no, no, it's not an issue, but there's some challenge there, and, but, I mean, you always did two papers, so I don't see why you would not do the third one, so. Yeah, but how do you feel if we have to extend it for the fall semester? I mean, it's all on you, I don't see any problem on my side, it's really about you.

I know it's not fun, but very often the last PhD year is the worst, and I think a lot of people, it's like that, but, you know, if I were you, I'm not trying to moralize you, but I would just focus on very short tasks that, you know, like just the goal of having some data that we can try stitching, not a whole data set or a large one, just like a very simple experiment with a simple shape, and starting from that, we will already have nice discussions, I'm sure, and it will lead us to somewhere interesting in the future. So it's just like, yeah, but I think it's pretty normal how you're feeling. Yeah.

I was feeling the same as well last year. Ah, okay, good. Yeah, no, totally, and, but I would really take Jean-Philippe's advice, just focus on small goals, like the idea is to break down, sometimes

we try to build the whole thing and then nothing works, and, you know, like for example, Dean collected data for like a month or two, and then when he started to validate it, he realized that the sensor broke or something broke halfway, and the point is build it, verify, validate it, so right now we can focus on like the stitching, automating the pipeline to take data, let's do a week on that, next week we talk, and then we try to understand more like different objects, and then we'll take more objects, and then we'll try to generate data.

At that point, we could try also to look, we'll start to get data, we'll start to try on ChatGPT, and don't get scared, it's super simple, I will help you on that, it's probably gonna, we'll build something and you will start it, you will go to sleep, you will have earned like \$200 of token during the night, and then we'll have the answer in the morning, so. Great. As long as you pay there, it's okay.

Great. But yeah, I will generate the key, and we'll take the bill. Great, thank you, I appreciate it.

Thank you so much. So, okay, great, so let's focus on that, let's meet next week, I, like, Tuesday would actually be a great day, because I'm gonna be working remotely from Italia, but I'm leaving Wednesday, so Wednesday will be in the airplane, like, yeah, and Thursday, so Tuesday would be great. Awesome, yeah, me too, it works.

Works for me, is it, like, do we have it at the same time, 11 AM, or? Doesn't matter for me, I will, yeah. It could be, just, yeah, it could be, just send us an invite at 11. It will be 5 PM for you, I think.

No, no, it'll be, I will still be there, in Quebec, it's, I go Wednesday. Okay, okay, okay. So next, the Tuesday after, if we meet, so I'm gonna be working, I'm not gonna be on vacation, but next Tuesday, it's, maybe we could do it a bit earlier, but yeah, 5 would be okay.

Yeah, great. Okay, I'll send the invite, thank you so much for your time, I appreciate your help, guidance, thank you so much. Thank you.

Take care. Thanks a lot. Bye.

Bye-bye, goodbye.

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